

Chapter 3

Intensity Transformation and Spatial Filtering

Part 3 Spatial Filtering

3.4 Fundamentals of Spatial Filtering

	Linear	nonlinear
smooth	Smooth Linear Filters	Smooth nonlinear Filters
Sharpening	Sharpening Linear filters	Sharpening nonlinear filters

3.4 Fundamentals of Spatial Filtering

$$g(x, y) = \sum_{m=-M}^M \sum_{n=-N}^N f(x+m, y+n) \cdot \text{mask}(m, n)$$

```
float r = 0;
for (int m=-M;m<=M;m++)
    for (int n=-N;n<=N;n++)
    {
        r+=f[x+m,y+n]*mask[M+m,N+n];
    }
g[x,y]=S(r);
```

Fundamentals of Spatial Filtering

*						
	5	5	4			
	4	7	4			
	4	4	3			
						#

0	-1	0
-1	5	-1
0	-1	0

*0	*-1	*0
*-1	*5	*-1
*0	*-1	*0

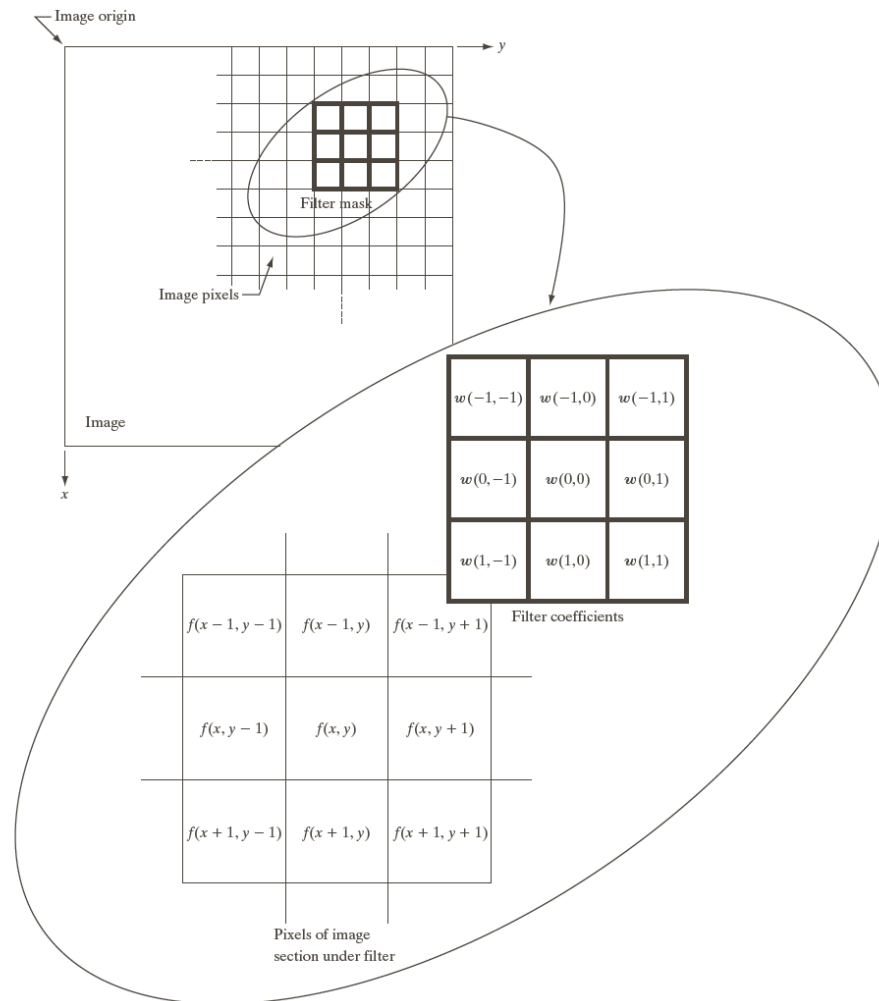


FIGURE 3.28 The mechanics of linear spatial filtering using a 3×3 filter mask. The form chosen to denote the coordinates of the filter mask coefficients simplifies writing expressions for linear filtering.

3.4 Fundamentals of Spatial Filtering

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

FIGURE 3.31

Another representation of a general 3×3 filter mask.

3.5 Smoothing Spatial Filters

- 3.5.1 Smoothing Linear Filters

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

$$\frac{1}{16} \times \begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

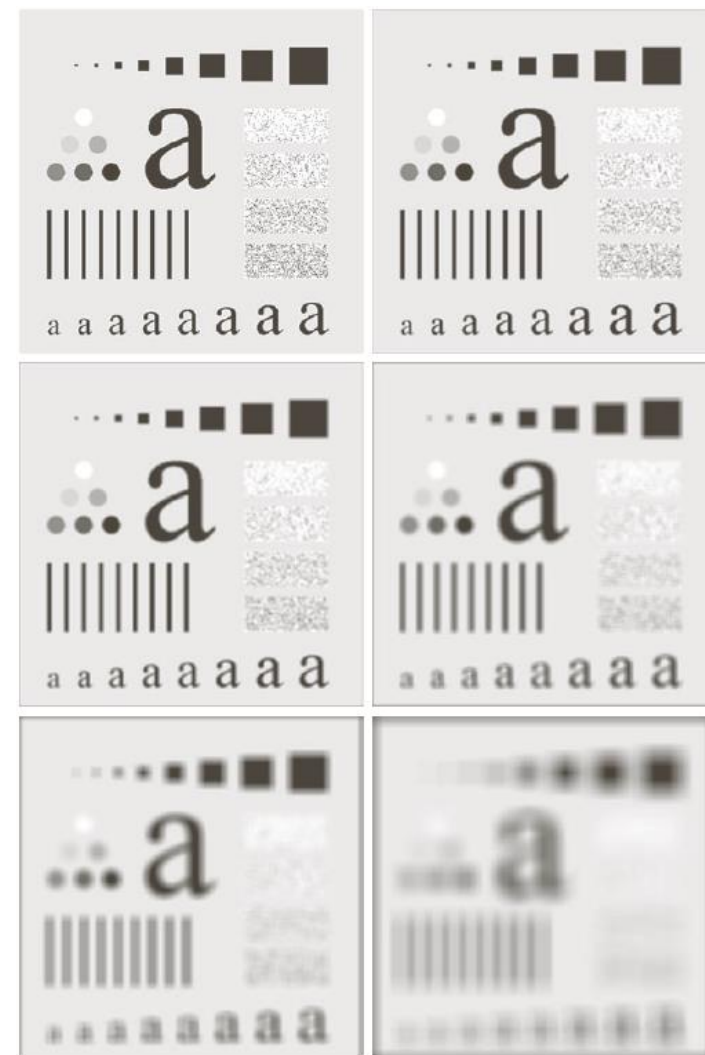
a b

FIGURE 3.32 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to 1 divided by the sum of the values of its coefficients, as is required to compute an average.

3.5.1 Smoothing Linear Filters

FIGURE 3.33 (a) Original image, of size 500×500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $m = 3, 5, 9, 15$, and 35 , respectively. The black squares at the top are of sizes $3, 5, 9, 15, 25, 35, 45$, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% black in increments of 20% . The background of the image is 10% black. The noisy rectangles are of size 50×120 pixels.

a	b
c	d
e	f



Directional Blur

0.2	0.2	0.2	0.2	0.2
-----	-----	-----	-----	-----

0.2
0.2
0.2
0.2
0.2

0.2				
	0.2			
		0.2		
			0.2	
				0.2

3.5.1 Smoothing Linear Filters

- [Programming of average filter]
- **Gaussian Blur**

$$\text{Mask}(x,y) = G(x,y) = \exp(-(x^2 + y^2)/(2 * r * r))$$

How to make a Gaussian Filter Mask

(1) Sampling: $G(x,y) = G[m,n]$;

(2) Normalized

$$s = \sum G[m,n], \quad G[m,n] = G[m,n]/s$$

[Programming of Gaussian Blur]

3.5.2 Order-Statistic (Nonlinear) Filters

- Median Filter
 - Median Filters are particularly effective in the presence of impulse noise (also called salt-and-pepper noise)
- Salt and pepper noise
 - (1) Random location;
 - (2) Independent value;
 - [Programming of Salt and pepper noise]

Salt and pepper



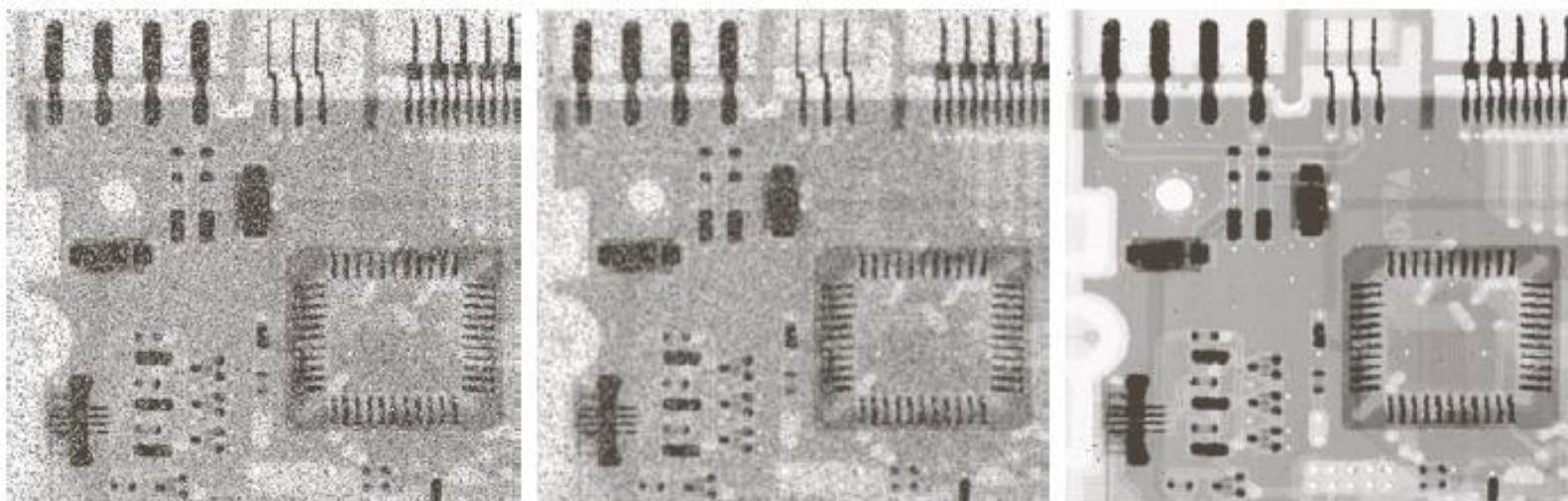
3.5.2 Order-Statistic (Nonlinear) Filters

- Median Operation
 - Median { 3,2,1,5,4 } = ?

{3,2,1,5,4} =>(queue)=> {1,2,3,4,5} => 3

[Programming of Median Filter]

3.5.2 Order-Statistic (Nonlinear) Filters



a b c

FIGURE 3.35 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

3.6 Sharpening Spatial Filters

- 3.6.1 Foundation
 - 1D: derivative f sub x .

$$f'(x) = f(x+1) - f(x)$$

$$f''(x) = f(x+1) + f(x-1) - 2f(x)$$

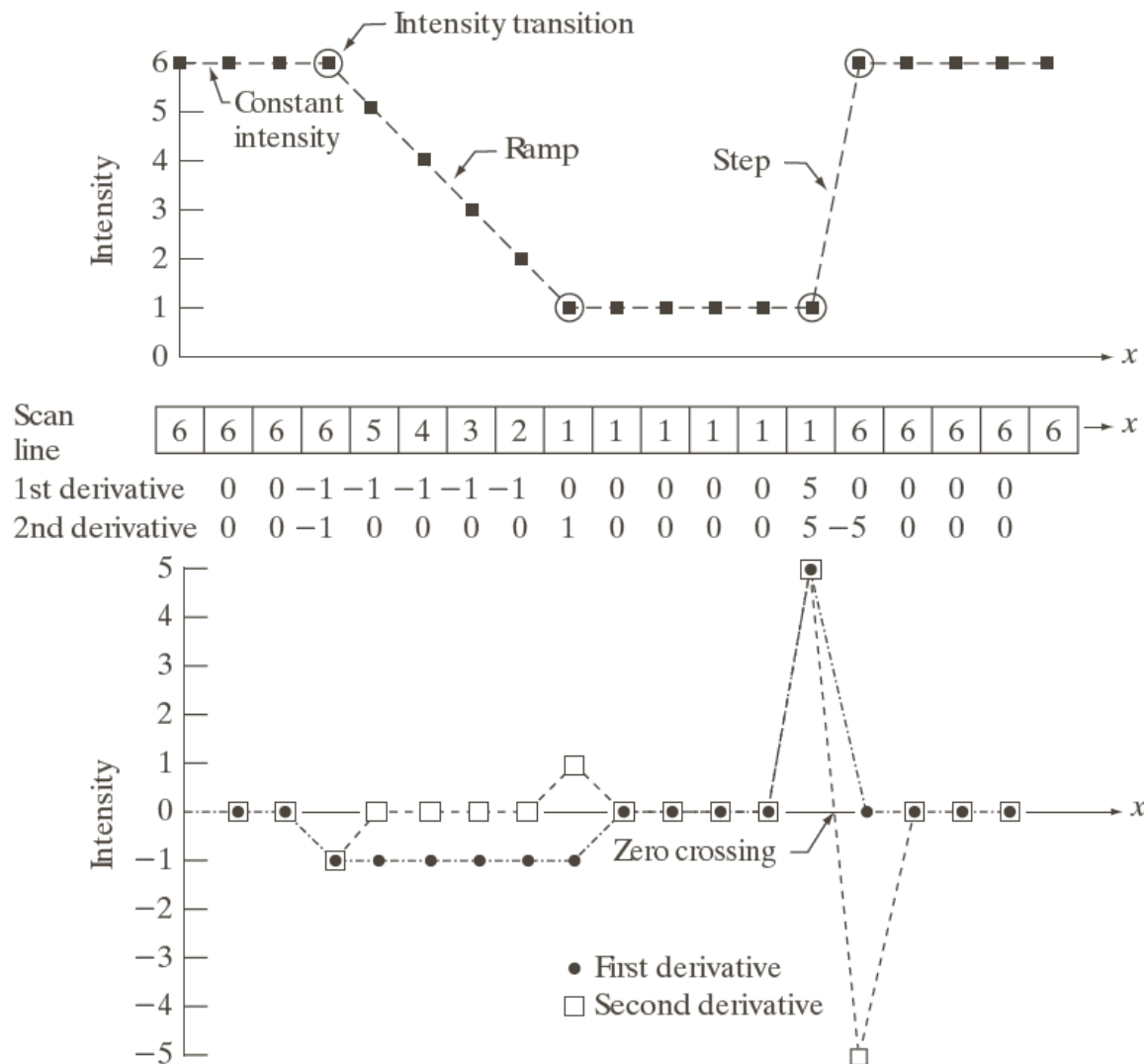
First derivative

- Must be zero of the area of constant intensity;
- Must be nonzero at the onset of an intensity step or ramp
- Must nonzero along ramps.

Second derivative

- Must be zero in constant area
- Must be nonzero at the onset and end of an intensity step or ramp
- Must be zero along ramps of constant slope

3.6.1 Foundation of Sharpening Spatial Filters



a
b
c

FIGURE 3.36

Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

3.6 Sharpening Spatial Filters

- 2D:
 - G_x : partial derivative f sub x .
 - G_y : partial derivative f sub y

$$\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

3.6.2 Laplacian (Second Derivative)

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

$$\frac{\partial^2 f}{\partial y^2} = f(y+1) + f(y-1) - 2f(y)$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

3.6.2 Laplacian (Second Derivative)

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a	b
c	d

FIGURE 3.37

(a) Filter mask used to implement Eq. (3.6-6).

(b) Mask used to implement an extension of this equation that includes the diagonal terms. (c) and (d) Two other implementations of the Laplacian found frequently in practice.

3.6.2 Laplacian (Second Derivative)

input	4	4	4	7	7	7	7
1 -2 1		0			0	0	
-1 2 -1		0			0	0	

High Pass filter

- [Programming of high pass filter]

$$\sum \sum Laplacian(m, n) = 0$$

3.6.2 Laplacian (Second Derivative)

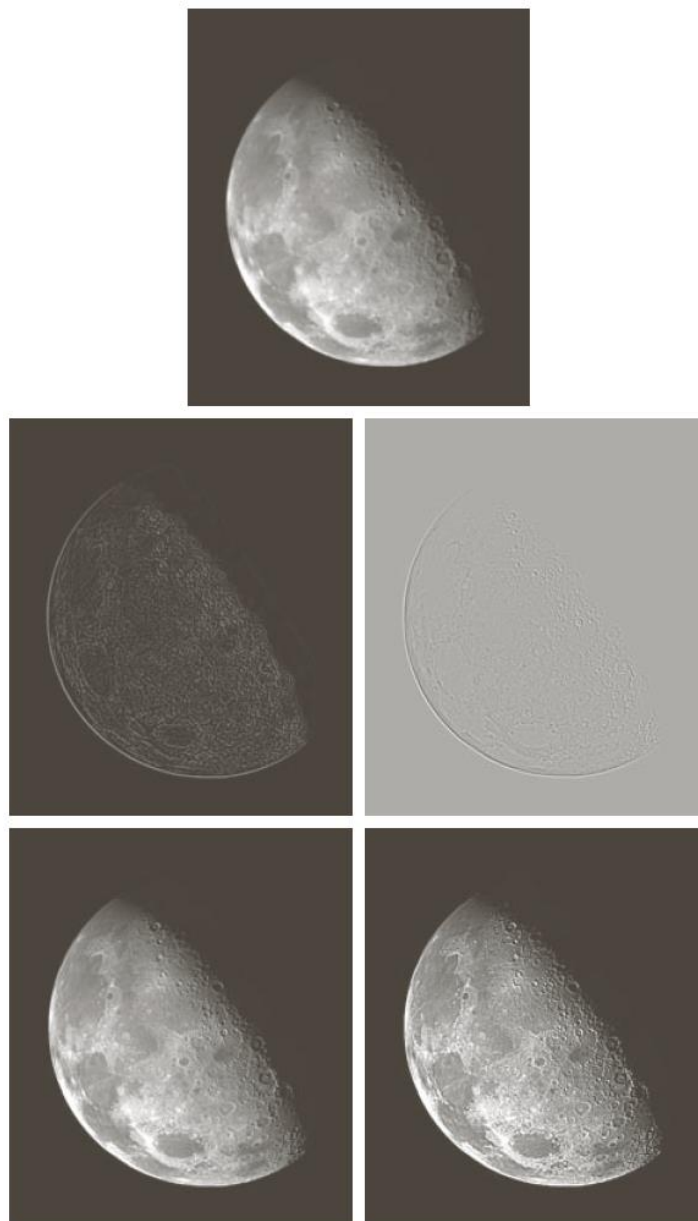
- [Programming of Laplacian]
- Sharpening Filter

$$g(x, y) = f(x, y) + c[\nabla^2 f(x, y)]$$

3.6.2 Laplacian (Second Derivative)

	$-c$	
$-c$	$1+4*c$	$-c$
	$-c$	

- [Programming of Sharpening Filter]



a
b c
d e

FIGURE 3.38

(a) Blurred image of the North Pole of the moon.

(b) Laplacian without scaling.

(c) Laplacian with scaling.

(d) Image sharpened using the mask in Fig. 3.37(a).

(e) Result of using the mask in Fig. 3.37(b).

(Original image courtesy of NASA.)

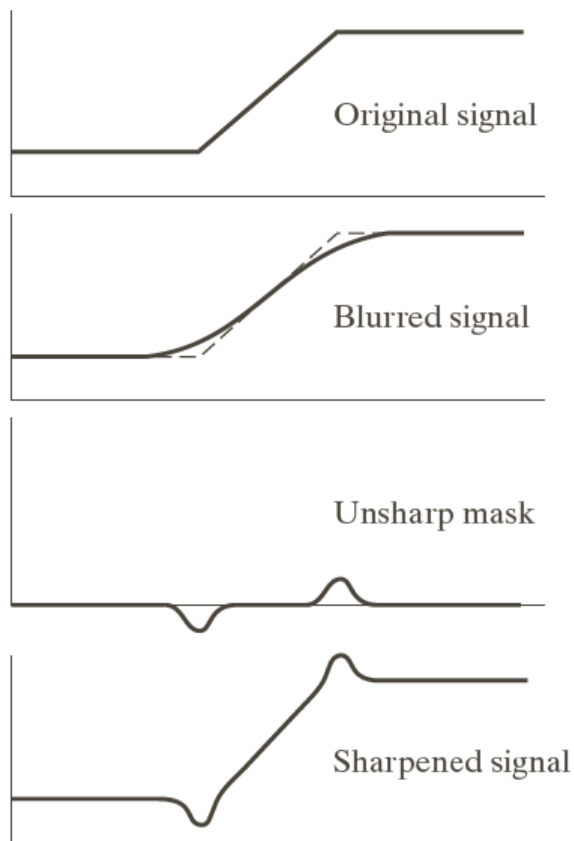
3.6.3 Unsharp Masking and Highboost Filtering

- Unsharp Masking
 - (1) Blur the original Image
 - (2) Subtract the blurred image from the original (the resulting difference called the mask)
 - (3) Add the mask to the original

$$g_{mask}(x, y) = f(x, y) - \bar{f}(x, y)$$

$$g(x, y) = f(x, y) + k \cdot g_{mask}(x, y)$$

3.6.3 Unsharp Masking and Highboost Filtering



a
b
c
d

FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).

3.6.3 Unsharp Masking and Highboost Filtering



a
b
c
d
e

FIGURE 3.40

(a) Original image.
(b) Result of blurring with a Gaussian filter.
(c) Unsharp mask.
(d) Result of using unsharp masking.
(e) Result of using highboost filtering.

3.6.4 Using the First-Order Derivatives for (Nonlinear) Image Sharpening –The Gradient

$$\nabla f = \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

$$M(x, y) \approx |g_x| + |g_y| \approx \max(|g_x|, |g_y|)$$

$$\phi(x, y) = a \tan 2(g_y(x, y), g_x(x, y))$$

Gradient

- Magnitude of Gradient

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

Gradient

- Approximate of magnitude (suitable computationally)

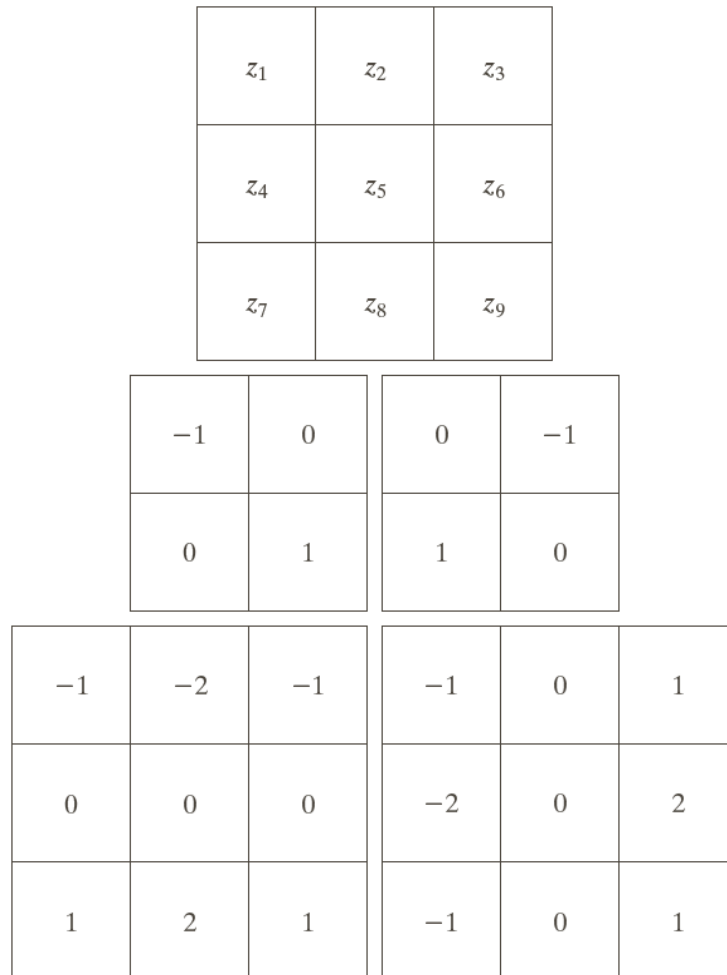
$$M(x, y) \approx |g_x| + |g_y| \approx \max(|g_x|, |g_y|)$$

First-Order Derivatives

- Question: max value of

$$f'_\theta(x, y) = Gx(x, y) \cos(\theta) + Gy(x, y) \sin(\theta)$$

3.6.4 Using the First-Order Derivatives for (Nonlinear) Image Sharpening –The Gradient



a
b c
d e

FIGURE 3.41

A 3×3 region of an image (the z s are intensity values).

(b)–(c) Roberts cross gradient operators.

(d)–(e) Sobel operators. All the mask coefficients sum to zero, as expected of a derivative operator.

Sobel mask

- G_x

-1	0	1
-2	0	2
-1	0	1

- G_y

-1	-2	-1
0	0	0
1	2	1

Y axis

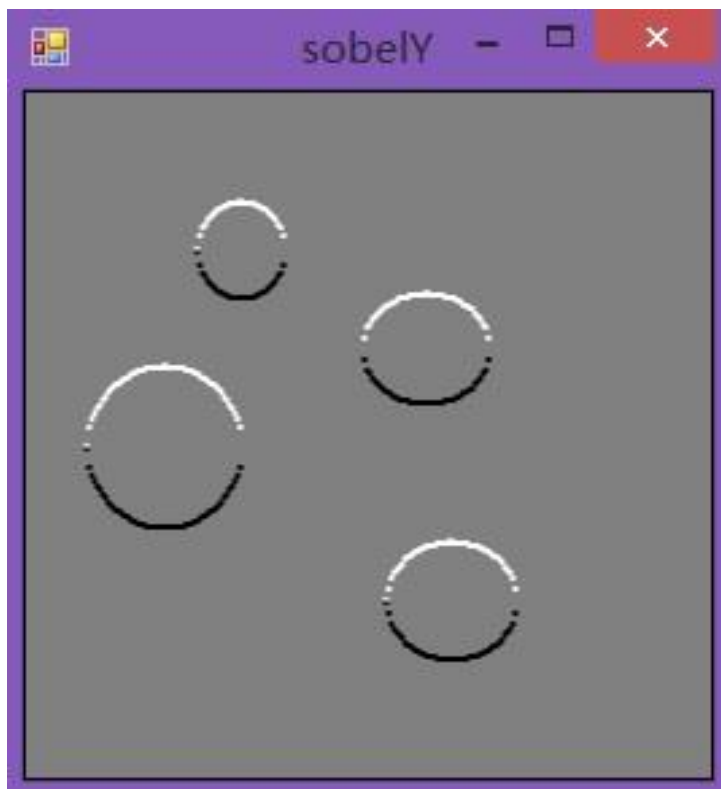


Programming of Sobel

relief effect

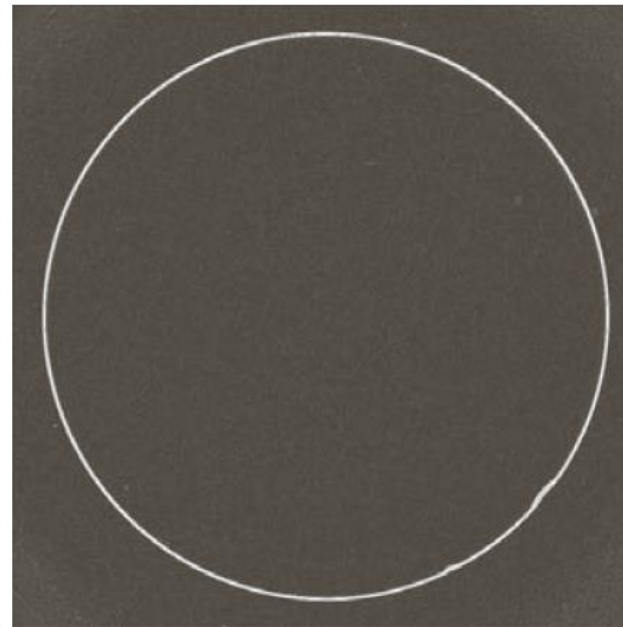
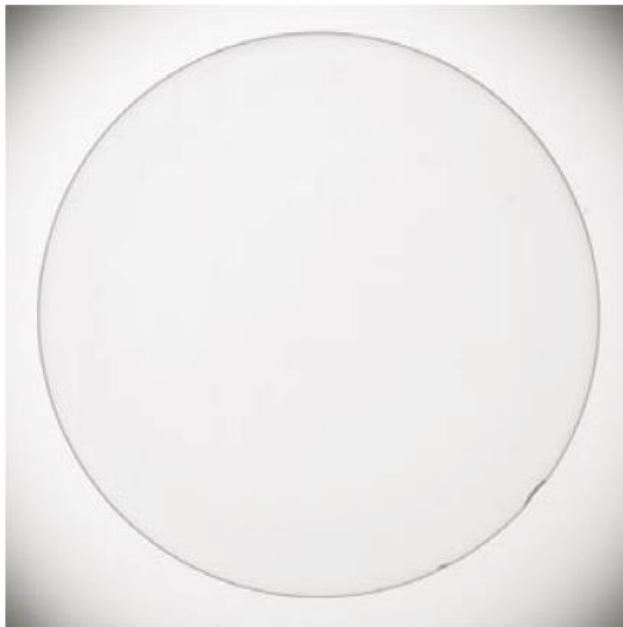


Sunken or Raised?



3.6.4 Using the First-Order Derivatives for (Nonlinear) Image Sharpening –The Gradient

- [Programming]



a b

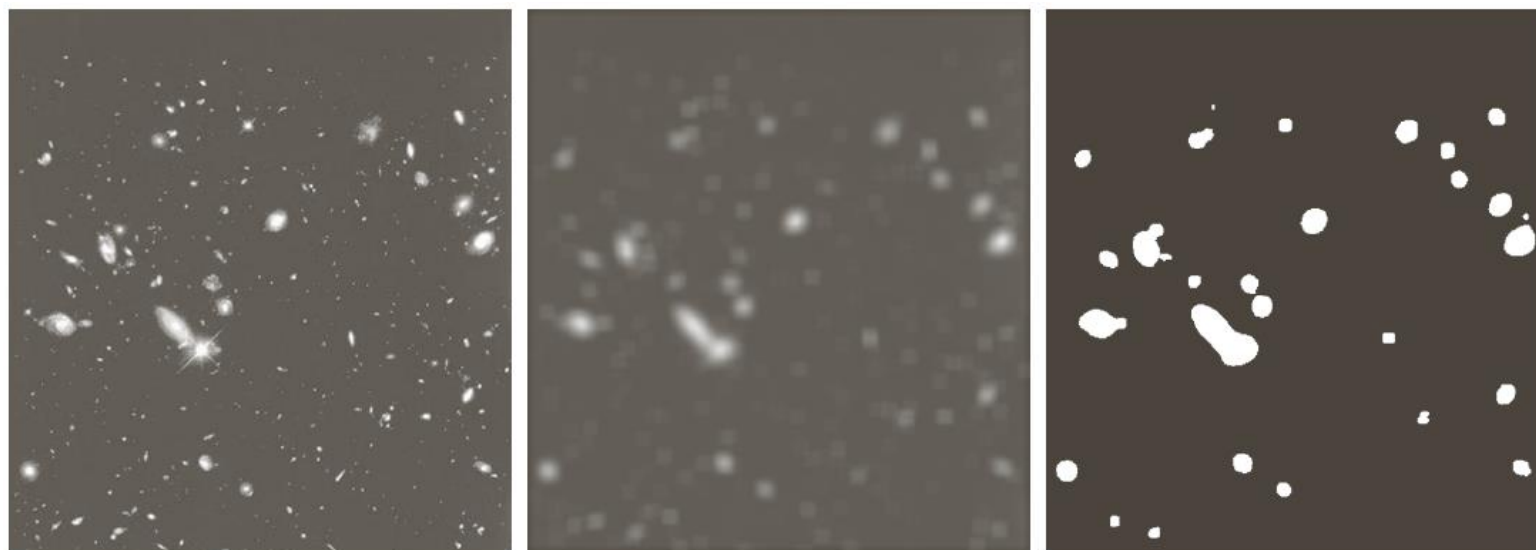
FIGURE 3.42

(a) Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock).

(b) Sobel gradient.

(Original image courtesy of Pete Sites, Perceptics Corporation.)

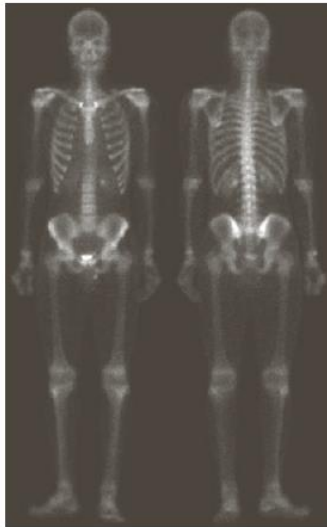
3.7 Combining Spatial Enhancement Methods



a b c

FIGURE 3.34 (a) Image of size 528×485 pixels from the Hubble Space Telescope. (b) Image filtered with a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

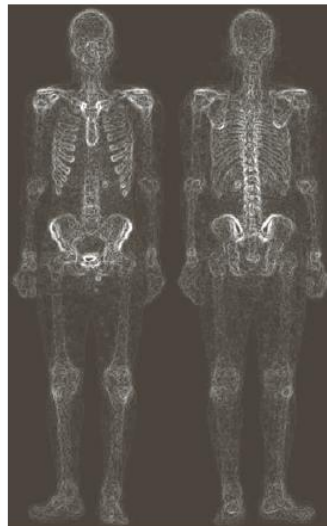
3.7 Combining Spatial Enhancement Methods



a b
c d

FIGURE 3.43

(a) Image of whole body bone scan.
(b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b).
(d) Sobel gradient of (a).

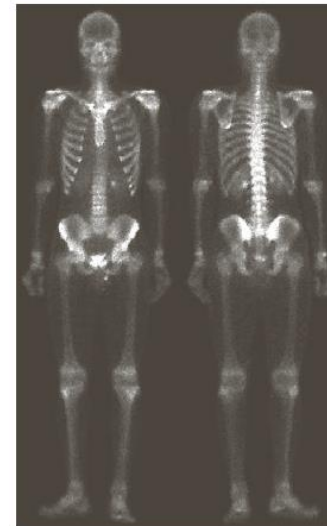
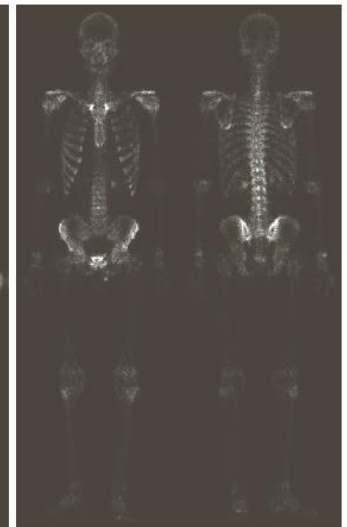
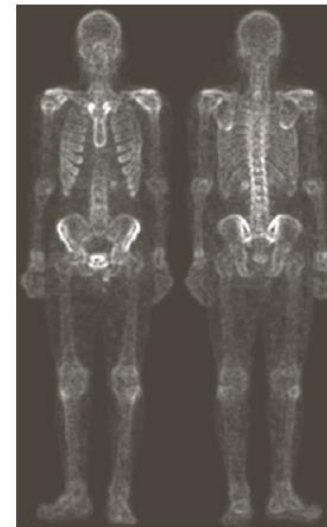


e f
g h

FIGURE 3.43

(Continued)

(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e).
(g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)



Algorithm Develop

